

Ruiyu Wang



rwan388@g.ucla.edu | 470-905-8972 | Los Angeles, CA

EDUCATION

University of California, Los Angeles <i>M.S. in Computer Science (Expected June 2028)</i>	09/2026 – 06/2028 Los Angeles, CA
Emory University <i>B.S. Computer Science; B.S. Applied Mathematics & Statistics; GPA: 3.93/4.00</i>	08/2022 – 05/2026 Atlanta, GA

RESEARCH & ENGINEERING EXPERIENCE

MagicLab Robotics <i>Research Intern</i>	07/2026 – 09/2026 Suzhou, China
Long-Term Memory for VLA-Based Robotic Manipulation	Summer 2026
– Surveyed memory mechanisms for long-horizon manipulation and participated in early design discussions on selective retrieval of task-relevant evidence from interaction history. – Implemented temporal/keyframe and object-level spatial supervision with offline label-generation tools for training memory retrieval components. – Helped establish training/evaluation workflows, ran selected model-training and testing experiments, and analyzed early memory behavior and failure modes.	
Department of Mathematics, Emory University <i>Undergraduate Research Assistant; Supervisor: Dr. Yuanzhe Xi</i>	09/2025 – Present Atlanta, GA
Reinforcement Learning for Multigrid Tuning (HYPRE)	2025 – Present
– Formulated mixed-type BoomerAMG configuration as a reinforcement-learning problem over strength thresholds, smoothing, coarsening, and relaxation choices. – Implemented policy-based tuners that adapt multi-parameter configurations across solves, avoiding Bayesian GP-based pre-tuning. – Benchmarked contextual bandits and episodic RL against GPTune/GPTuneBand to assess tuning efficiency and stability.	
Emory Graph Mining Lab, Department of Computer Science <i>Undergraduate Research Assistant; PI: Dr. Carl J. Yang</i>	08/2024 – 12/2025 Atlanta, GA
AceSearcher: Reinforced Self-Play for Reasoning and Search	03/2025 – 05/2025
– Co-developed a self-play RAG framework in which one LLM alternates as Decomposer, breaking down multi-hop queries, and Solver, integrating retrieved evidence. – Benchmarked the framework across question-answering tasks with standardized prompting of baseline LLMs. – Co-authored the work as fourth author; accepted to NeurIPS 2025 as a Spotlight.	
Knowledge Graph-Augmented LLMs for Disease Prediction	05/2025 – 12/2025
– Led an end-to-end KG-guided disease-prediction pipeline integrating structured reasoning chains into LLaMA-8B fine-tuning on 1,000 MIMIC-III visits. – Achieved AUC/AUPR comparable to traditional ML baselines and maintained performance with only 400 patient visits, demonstrating data efficiency. – Led the project and authored the first-author study; accepted to AMIA 2026.	
Hypergraph Pre-training for Atrial Fibrillation Prediction	08/2024 – 02/2025
– Developed hypergraph-transformer pre-training pipelines on a 7,780-visit stroke cohort and transferred 32-D representations to an ESUS cohort ($n = 510$). – Integrated clinical features with learned embeddings, improving AUROC by 5–15% over from-scratch baselines. – Co-authored the work as third author; accepted to AMIA 2026.	
MedAssist: LLM Assistant for Electronic Health Records	01/2025 – 04/2025
– Built a retrieval-augmented assistant that combines external medical knowledge with tabular EHR evidence through an LLM agent, retriever, and SQL/tool executors. – Converted medical literature into structured knowledge graphs and designed pair, direction, and type-variance metrics; third-author work accepted to the WWW 2025 Demo Track.	
Department of Computer Science, Emory University <i>Undergraduate Research Assistant; PI: Dr. Jiaying Lu</i>	12/2024 – 11/2025 Atlanta, GA

BioMedJImpact: Biomedical Journal-Impact Analysis

12/2024 – 11/2025

- Built a dataset of 1.74M PMC articles across 2,744 biomedical journals and a three-stage LLM pipeline to identify AI-related research and assign subfields, supported by human evaluation.
- Analyzed collaboration, AI engagement, and citation impact; led the project and authored the first-author work presented at the KDD Undergraduate Consortium 2026.

INDUSTRY EXPERIENCE

Tongcheng Holdings Limited

06/2025 – 08/2025

AI Engineer Intern

China

- Fine-tuned a speaker-diarization backbone and QMF, reducing EER from 0.87% to 0.37% and improving internal recall from 79% to 96%.
- Designed an evaluation framework for the DeepTrip LLM travel assistant covering factual accuracy, constraint adherence, and response clarity.

JINGDONG (JD)

05/2025 – 06/2025

AI Intern

China

- Fine-tuned LoRA adapters for brand-aligned mascot image generation and documented the resulting workflow for a non-AI team.

PUBLICATIONS & PRESENTATIONS

Xu, R., Zhuang, Y., Dong, Z., **Wang, R.**, Yu, Y., Ho, J. C., Zhang, L., Wang, H., Shi, W., & Yang, C. (2025). “AceSearcher: Bootstrapping Reasoning and Search for LLMs via Reinforced Self-Play.”  *Advances in Neural Information Processing Systems (NeurIPS 2025)*, Spotlight.

Wang, R., Vinh, T., Xu, R., Zhou, Y., Lu, J., Pasquel, F., Ali, M. K., & Yang, C. (2026). “Knowledge Graph Augmented Large Language Models for Disease Prediction.”  *AMIA 2026 Informatics Summit*.

Xie, Y., Wu, Y., **Wang, R.**, Nahab, F., Hu, X., & Yang, C. (2026). “Enhanced Atrial Fibrillation Prediction in ESUS Patients with Hypergraph-based Pre-training.”  *AMIA Joint Summits on Translational Science Proceedings*.

Xu, R., Shi, W., **Wang, J.**, Zhou, J., & Yang, C. (2025). “MedAssist: LLM-Empowered Medical Assistant for Assisting the Scrutinization and Comprehension of Electronic Health Records.”  *Companion Proceedings of the ACM Web Conference 2025*, Demo Track.

Wang, R., Xie, Y., Hu, X., Yang, C., & Lu, J. (2026). “BioMedJImpact: A Comprehensive Dataset and LLM Pipeline for AI Engagement and Scientific Impact Analysis of Biomedical Journals.”  arXiv preprint; KDD Undergraduate Consortium 2026.

LEADERSHIP & SERVICE

Emory University Mathematics Association

Fall 2025 – Spring 2026

Vice President

Atlanta, GA

- Co-organized research, graduate-school, and major/minor panels by coordinating faculty and student speakers and managing event logistics.
- Advised MCM and Kaggle teams on team formation, modeling strategy, evaluation, and report writing; led preparation workshops and mock judging sessions.

HONORS & AWARDS

Honorable Mention, Mathematical Contest in Modeling

Spring 2025

Dean’s List, Emory College of Arts and Sciences

Fall 2023 – Fall 2024

Best Team and Insight Award, DataFest, Emory University

Spring 2024

TECHNICAL SKILLS & INTERESTS

Programming & Systems: Python, C, Java, Linux, Git, Docker, ARM Assembly

Machine Learning: PyTorch, deep learning, reinforcement learning, model training and evaluation, LLM fine-tuning

Robotics: ROS 2 (basic), kinematics (basic), motion planning (basic), feedback control (basic)

Embedded Systems: STM32 (HAL), UART, timers, PWM, interrupts, ADC, PWM-based motor/servo control

Other Interests: Guitar, music composition, badminton

RELEVANT COURSEWORK

Fall 2026: MECH&AE C263A – Kinematics of Robotic Systems; EC ENGR M240A – Linear Dynamic Systems

Machine Learning: Machine Learning, Deep Learning, Natural Language Processing, Data Mining, Probabilistic Machine Learning

Mathematics & Statistics: Linear Algebra, Mathematical Statistics, Numerical Analysis